

Project Synopsis

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- **Title**
 - **Sentiment Analysis of Airline Social Media Data for Proactive Brand Management**

- **Problem Statement (200 words)**
 - The modern aviation industry operates in a highly visible digital landscape where platforms like Twitter serve as a real-time, public-facing customer service desk. When passengers experience flight delays, lost luggage, or logistical failures, their frustration frequently spills over into the public domain, causing immediate and measurable brand damage. However, raw social media data is inherently unstructured, noisy, and generated at a velocity that makes manual analysis completely impossible at scale. Millions of interactions are posted daily across various timezones, overwhelming traditional customer support teams and burying critical operational insights under an avalanche of unstructured conversation.
 - The core problem the industries are facing is that many brands lack the automated, intelligent mechanisms required to instantly categorize this massive volume of feedback. Without an automated machine learning pipeline, it is incredibly difficult to identify whether the primary cause of negative sentiment is purely logistical (for e.g., unavoidable weather delays, mechanical issues) or organizational (for e.g., poor customer service response, unhelpful staff, long phone hold times). Without a structured, mathematical way to process this unstructured text, companies struggle to prioritize their crisis management efforts, allocate resources efficiently, and resolve the root causes of customer dissatisfaction before they escalate into highly damaging, viral PR crises.

- **Objectives of the Project (3objectives)**
 - **To analyze and extract actionable patterns from unstructured social media data:** The primary objective of this research is to utilize Natural Language Processing (NLP) techniques to analyze a historical dataset of airline customer tweets. By conducting exploratory data analysis and bigram frequency mapping, the study aims to mathematically identify the most

frequent operational triggers (such as flight cancellations) and map how they directly correlate with secondary institutional complaints (such as poor customer service).

- **To establish the strategic significance of text analytics in brand management:** The second objective is to prove the business value of deploying supervised machine learning on raw customer feedback. The significance of this chosen project lies in demonstrating that automated text classification can instantly uncover specific organizational bottlenecks, allowing companies to prioritize their crisis management efforts and allocate resources far more efficiently than manual analysis would permit.
- **To develop and deploy a functional machine learning pipeline:** The final objective, and the primary contribution of the learner to this field of study, is the end-to-end engineering of a predictive classification model. This involves training algorithms to accurately predict sentiment and deploying the optimized model via a local Flask web application, thereby creating a tangible, automated tool capable of categorizing new text inputs for proactive brand management.

- **Project Methodology (500 words)**

- **Brief Overview of the Methodology :** This project relies on Natural Language Processing (NLP) and supervised machine learning to mathematically classify unstructured text data into distinct emotional categories: positive, negative, and neutral. The methodology follows an end-to-end data science lifecycle, transitioning systematically from raw data ingestion to a fully deployed predictive model. The pipeline begins with rigorous data cleaning, utilizing regular expressions to remove social media noise such as URLs, special characters, and user mentions. This is followed by advanced text preprocessing, where raw sentences are broken down through tokenization. Common filler words are eliminated via stop-word removal, and words are normalized to their root forms using lemmatization to account for frequent social media misspellings. Once standardized, the methodology moves into feature engineering. The text is transformed into numerical formats using vectorization techniques like TF-IDF (Term Frequency-Inverse Document Frequency), which mathematically weighs the statistical importance of specific vocabulary. Finally, the methodology concludes with training robust classification algorithms, such

as Naive Bayes and Random Forest. Because the dataset features a severe class imbalance, the models are carefully evaluated using advanced metrics including Precision, Recall, and the F1-score to ensure reliable and unbiased predictions.

- **Type of Project (Research and Application-Based)** : This project is strategically designed as a hybrid, encompassing both research-based and application-based methodologies. The initial phase is heavily research-based, focusing on Exploratory Data Analysis (EDA) to uncover hidden lexical patterns and mathematical correlations within the raw text. Rather than making assumptions, this research phase objectively maps out operational failure points—such as identifying the most frequent two-word complaint phrases and proving how physical tweet character length correlates heavily with negative sentiment. Following this research, the study transitions into a practical, application-based project. The insights gathered are mathematically applied to train supervised machine learning algorithms. The ultimate goal of this application-based phase is not just to analyze historical data, but to engineer a tangible, highly scalable software solution for modern brand management. The final classification model is serialized and integrated into a local web application using the Flask framework, demonstrating how businesses can actively deploy artificial intelligence to categorize real-time text inputs.
- **Data Collection Methods (Secondary)** : The project relies strictly on secondary data collection, bypassing the need for real-time web scraping or manual Twitter API extraction. The research utilizes an existing, highly structured dataset consisting of over 14,000 individual tweets directed at six major United States airlines during early 2015. This specific secondary dataset was selected because it provides a rich, multi-dimensional variety of features beyond just the raw text, allowing for much deeper operational insights. It includes the specific airline tagged, the user's geographical timezone, the day of the week, and categorized reasons for negative complaints, such as "Late Flight" or "Customer Service Issue." Most importantly, the dataset includes

human-annotated sentiment labels classifying each tweet as positive, negative, or neutral. Having this human-verified ground truth is an absolute necessity for supervised machine learning, as it provides the mathematically reliable baseline required to properly train and evaluate the predictive algorithms.

- **Limitation (200 words)**

- **Sarcasm and Irony Detection:** Traditional machine learning algorithms often struggle to accurately classify sarcastic statements, such as a passenger tweeting, "Thanks a lot for losing my luggage, great job!" Because this study relies on foundational classification models like Random Forest and Naive Bayes, developing deep-learning contextual algorithms to detect complex irony will not be addressed.
- **Multilingual Analysis:** The global aviation industry handles customer feedback in dozens of languages. However, the text preprocessing, tokenization, and stop-word removal techniques utilized in this pipeline are strictly configured for the English language. The complexities of translating or analyzing non-English social media data fall outside the scope of this project.
- **Multimedia Processing:** Modern social media complaints frequently include attached images or videos of delayed flight boards and damaged luggage. This study is strictly limited to text-based NLP; therefore, computer vision or audio processing to analyze attached multimedia files will not be addressed.
- **Real-Time Data Integration:** Finally, this project focuses entirely on static batch processing and local web deployment. Consequently, the research will not address the backend engineering complexities of real-time streaming data ingestion or live API integration with current social media platforms.

- **Work Plan (Week 1 to Week 8)**

Week No.	Activities Completed
Week 1	a) Conduct a comprehensive literature review of existing NLP sentiment analysis methodologies. b) Establish the technical foundation by configuring the local Python development environment. c) Set up virtual environments and install necessary data science dependencies (e.g., Pandas, Scikit-Learn).
Week 2	a) Ingest the raw secondary dataset and perform initial data cleaning to handle missing values. b) Conduct preliminary Exploratory Data Analysis (EDA) to visualize operational failure factors. c) Map the volume of negative sentiment across specific airlines and temporal factors (timezones).
Week 3	a) Generate advanced NLP visualizations, including distinct Word Clouds for positive and negative lexicons. b) Mathematically identify the most common operational complaints via Bigram frequency analysis. c) Verify the severity of the class imbalance within the dataset using distribution visualizations.
Week 4	a) Remove social media-specific noise from the unstructured text (URLs, emojis, special characters). b) Perform core NLP preprocessing, including text tokenization and stop-word removal.

	c) Transform the cleaned text into numerical features using vectorization techniques (e.g., TF-IDF).
Week 5	a) Transform the cleaned text into numerical features using vectorization techniques (e.g., TF-IDF). b) Train multiple supervised classification algorithms (such as Naive Bayes and Random Forest). c) Compare the baseline predictive accuracy of the different machine learning algorithms.
Week 6	a) Evaluate the models using advanced metrics, specifically focusing on Precision, Recall, and the F1-score. b) Conduct hyperparameter tuning on the best-performing model to optimize performance and reduce overfitting. c) Serialize and export the final optimized mathematical model (via Pickle) for future deployment.
Week 7	a) Develop the front-end user interface (HTML/CSS) for the local web application. b) Build the back-end routing system using the Flask framework to handle incoming user data. c) Integrate the serialized NLP classification model into the web application backend.
Week 8	a) Conduct end-to-end integration testing of the Flask application to ensure model stability. b) Formally compile and format the final research paper according to university standards. c) Prepare the final project documentation and presentation for official submission.